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GitHub Link	https://github.com/192425292simats-byte/MLA0403-DEEP-LEARNING

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING
SLOT :A

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I (V Sunil Kumar / 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: V Sunil

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Total						

Signature of the Faculty:

Total Marks :

Q1. CNN for Automated Medical Image Classification — 10 Marks

1. Suitable CNN Architecture

For classifying chest X-ray images into **Normal** and **Disease-Affected**, a CNN can be designed as follows:

Input Image → **Convolution** → **ReLU** → **Pooling** → **Convolution** → **ReLU** → **Pooling** → **Convolution** → **ReLU** → **Global Average Pooling/Flatten** → **Fully Connected Layer** → **Dropout** → **Output**

A suitable architecture is:

- **Input:** Resize X-ray images to $224 \times 224 \times 1$ because X-rays are grayscale.
- **Convolution Layer 1:** 32 filters of size 3×3 to detect basic edges and textures.
- **ReLU activation:** Introduces non-linearity.
- **Max Pooling:** 2×2 to reduce spatial dimensions.
- **Convolution Layer 2:** 64 filters to learn more complex patterns.
- **ReLU + Max Pooling**
- **Convolution Layer 3:** 128 filters to identify higher-level structures and abnormal regions.
- **Global Average Pooling or Flattening:** Converts feature maps into features suitable for classification.
- **Fully Connected Layer:** Combines learned features.
- **Dropout:** Reduces overfitting.
- **Output Layer:** One neuron with **sigmoid activation** for binary classification.

The model can be trained using **binary cross-entropy loss** and an optimizer such as Adam.

2. Motivation and Role of CNN Layers

CNNs are appropriate because medical images contain spatial patterns such as **edges, textures, shapes and abnormal regions**. The case specifically identifies the need to automatically learn these visual features rather than manually designing them.

Layer	Purpose
Input	Receives standardized X-ray images.
Convolution	Extracts local visual features using learnable filters.
Filters/Kernels	Detect patterns such as edges, lines, textures and abnormal structures.
ReLU	Adds non-linearity and allows the network to learn complex patterns.
Pooling	Reduces feature-map size and computational requirements while retaining important information.
Fully	Combines extracted features for final classification.

Layer	Purpose
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Connected	
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Output	Produces probability of normal/disease-affected classification.
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Initially, filters learn **low-level features** such as edges and intensity changes. Deeper layers combine these into **higher-level features**, such as lung structures and possible disease-related patterns.

3. Parameter Sharing

Parameter sharing is one of the major advantages of CNNs.

In a fully connected network, every neuron may connect to every pixel. For example, a 224×224 grayscale image contains:

$$224 \times 224 = 50,176 \text{ pixels}$$

A fully connected layer with 1,000 neurons would require approximately:

$$50,176 \times 1,000 = 50,176,000 \text{ weights}$$

In contrast, a 3×3 CNN filter has only:

$$3 \times 3 = 9 \text{ weights} + 1 \text{ bias}$$

The same filter is **shared across different image locations**.

Therefore, instead of learning separate parameters for every location, CNNs learn a small set of filters that can detect the same feature anywhere in the image.

Benefits:

- Significantly fewer trainable parameters.
- Lower memory requirement.
- Reduced computational cost.
- Better ability to detect features at different spatial locations.
- Improved generalization.

This directly addresses the assessment requirement concerning how parameter sharing reduces trainable parameters compared with fully connected networks.

4. Regularization Techniques

The dataset contains variations in **size, brightness and image quality**, and a deep model can overfit the training data.

a) Data Augmentation

Training images can be randomly modified using:

- Small rotations
- Horizontal shifts
- Zooming
- Cropping
- Brightness adjustments

- Contrast adjustments

This increases the diversity of training examples and helps the model become robust to variations in X-ray acquisition.

However, augmentation must be medically appropriate. Excessive transformations should be avoided because they may create unrealistic anatomical structures.

b) Dropout

Dropout randomly disables a percentage of neurons during training.

For example:

Dense → Dropout(0.5) → Output

This prevents the network from depending too heavily on particular neurons and improves generalization.

c) Batch Normalization

Batch normalization normalizes intermediate activations during training.

Advantages include:

- More stable training.
- Faster convergence.
- Reduced sensitivity to parameter initialization.
- Some regularization effect.

Additional methods such as **early stopping, weight decay and class balancing** can also be considered.

5. AlexNet vs ResNet

Factor	AlexNet	ResNet
Architecture	Conventional CNN	CNN with residual/skip connections
Depth	Relatively shallow	Can be much deeper
Feature learning	Good	Stronger hierarchical feature learning
Training	Easier than very deep networks	Skip connections make deep training easier
Vanishing gradient	More manageable because it is shallower	Residual connections strongly reduce the problem
Computational requirement	Generally lower	Higher depending on ResNet version
Performance	Good baseline	Usually stronger for complex image tasks
Medical-image suitability	Possible	Generally more suitable

Recommendation

ResNet is more appropriate for this application, particularly a moderate version such as **ResNet-18 or ResNet-34**, because medical X-rays can contain subtle and complex patterns.

Residual connections allow information and gradients to pass through shortcut paths, making deeper networks easier to train. This supports stronger feature learning and potentially better classification performance.

For a healthcare organization, however, accuracy alone is insufficient. The model should also be evaluated using:

- Accuracy
- Precision
- Recall/Sensitivity
- Specificity
- F1-score
- ROC-AUC
- Confusion matrix

For disease screening, **sensitivity/recall is especially important**, because incorrectly classifying a disease-affected image as normal can have serious consequences.

Final choice: ResNet, because it provides stronger feature learning and deeper representation capability while residual connections make training practical. The architecture should be selected according to available computational resources. This directly addresses the case requirement to compare AlexNet and ResNet using feature learning, depth, computation and classification performance.

Q2. CNN-Based Autonomous Vehicle Object Recognition — 10 Marks

1. CNN Architecture for Autonomous Vehicles

The autonomous vehicle must recognize:

- Cars
- Pedestrians
- Traffic signs
- Bicycles
- Road obstacles

under different **lighting conditions, viewing angles, distances and road environments**.

A suitable CNN pipeline is:

Camera Image → Preprocessing → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Deep Convolution Blocks → Feature Representation → Classification/Detection Output

Example:

Input: 224×224×3 RGB image

→ Conv 32 filters

→ ReLU

→ Max Pooling

→ Conv 64 filters

→ ReLU

→ Max Pooling

→ Conv 128 filters

→ ReLU

→ Conv 256 filters

→ Global Average Pooling

→ Fully Connected Layer

→ Output

For a real autonomous vehicle, an object-detection architecture would generally be preferable to simple image classification because the system must identify **what objects are present and where they are located**.

2. Purpose of CNN Layers

Input Layer

Receives RGB camera images and standardizes their dimensions and pixel values.

Convolution Layer

Applies filters to local regions of the image.

Early filters may detect:

- Edges
- Corners
- Lines
- Simple textures

Deeper filters can detect:

- Wheels
- Vehicle shapes
- Human body structures
- Traffic-sign patterns
- Complex road objects

Activation Layer

ReLU introduces non-linearity:

$$\text{ReLU}(x) = \max(0, x)$$

This allows the network to model complex relationships.

Pooling Layer

Pooling reduces spatial dimensions.

For example, **2×2 max pooling** reduces the feature-map size while retaining strong activations.

This reduces computational requirements and provides some tolerance to small changes in object position.

Fully Connected/Output Layer

The learned features are combined to determine the object category.

For multiple classes, the final classification layer can use **softmax**.

For object detection, the output must additionally provide **bounding-box coordinates and class predictions**.

The assessment specifically requires explanation of how filters progressively learn low-level and high-level features.

3. Importance of Parameter Sharing

Suppose an image contains a car on the left side and another car on the right side.

A CNN does not need separate filters to learn the concept of a **car edge or wheel pattern** at every position.

The same filter is applied throughout the image.

For example:

Filter → Left region → Center region → Right region

The same learned weights detect similar features regardless of their location.

Therefore parameter sharing:

- Reduces the number of parameters.
- Reduces memory requirements.

- Reduces computation compared with a fully connected network.
- Allows the same visual feature to be detected at different locations.
- Improves translation-related feature detection.

This is particularly important for autonomous vehicles because objects may appear anywhere in the camera frame.

4. Overfitting and Regularization

The vehicle dataset may contain images from limited roads, cameras or weather conditions. If the CNN memorizes training images instead of learning general features, it may perform poorly in new environments.

Suitable techniques include:

1. Data Augmentation

Generate variations using:

- Brightness changes
- Contrast changes
- Cropping
- Scaling
- Small rotations
- Horizontal transformations where appropriate

This helps the model handle different road conditions.

2. Dropout

Randomly removes neurons during training and reduces dependence on specific features.

3. Batch Normalization

Stabilizes intermediate activations and supports efficient training.

4. Weight Decay

Penalizes excessively large model weights and improves generalization.

5. Early Stopping

Stops training when validation performance stops improving.

6. Diverse Training Data

Include different:

- Weather conditions
- Day/night conditions
- Camera viewpoints
- Object distances
- Road environments

This is especially important because the case explicitly states that the system must work under varying

lighting, viewing angles, distances and road environments.

5. AlexNet vs ResNet

Feature	AlexNet	ResNet
Basic design	Traditional deep CNN	Residual CNN
Depth	Around 8 learnable layers	Can have 18, 34, 50, 101+ layers
Skip connections	No	Yes
Vanishing gradient	More susceptible in deeper extensions	Significantly reduced
Feature learning	Good	Excellent for deeper hierarchical features
Training difficulty	Relatively straightforward	Deep networks are practical because of residual connections
Parameter efficiency	Large parameter count, especially in dense layers	Modern variants can achieve strong performance efficiently
Accuracy	Good baseline	Generally better for complex visual recognition
Real-time suitability	Lower computational burden	Depends on variant; lightweight ResNet versions can be suitable
Scalability	Limited compared with modern deep networks	Highly scalable

The assessment specifically asks for comparison in terms of architecture, depth, parameter efficiency, training difficulty, vanishing gradients and feature-learning capability.

6. Recommended Architecture

ResNet is the better choice for the autonomous vehicle system.

The main reasons are:

1. **Higher feature-learning capability:** Deeper layers can learn complex representations of pedestrians, vehicles, signs and obstacles.
2. **Residual connections:** Skip connections make it easier to train deeper networks and reduce degradation/vanishing-gradient problems.
3. **Robust recognition:** Deeper hierarchical features can improve recognition under different viewpoints and environments.
4. **Better scalability:** Different ResNet variants can be selected depending on hardware resources.
5. **Real-time consideration:** A smaller ResNet variant can be selected when latency is critical, while more powerful variants can be used when greater accuracy is required.

However, for an actual autonomous vehicle, the final system should not simply use a large ResNet without considering latency. **Accuracy and computational cost must be balanced.** The selected model should

meet the vehicle's real-time inference requirement.

Therefore, a **lightweight ResNet-based model** would be a strong choice when the system needs a balance between **accuracy, computational cost and real-time performance**.

The assessment explicitly requires the final recommendation to justify the architecture using these three factors.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
